

Individual Contribution Report

Design and Creative Technologies

Torrens University, Australia

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Subject Code: ISY 503

Subject Name: Intelligent Systems

Assessment No.: 3

Title of Assessment: Case Study Report

Lecturer: Dr. Nandini Sidnal

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1. Group Details

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- Source code: <https://github.com/lfariabr/review-pulse>
- Presentation: [ISY503 Faria L Assessment 3.pptx](#)
- Live Demo: <https://review-pulse.streamlit.app/>

2. Team Contribution Table

Member	Main contribution	%
Luis	Technical architecture, implementation, integration, testing, documentation, release management, and coordination	50%
Samiran	Problem framing, live demonstration support, and future improvement planning	25%
Victor	DistilBERT/model experimentation and integration support, dataset analysis, error analysis, ethics presentation	25%

3. Individual Report

My primary contribution to ReviewPulse was the full technical implementation and hardening of the ML pipeline - the modular Python package structure, the Streamlit web app, the automated test suite, and the project documentation.

On the data side, I built the pseudo-XML parser (*src/data/parser.py*), the preprocessing pipeline with negation expansion (*src/data/preprocess.py*), and the vocabulary and PyTorch DataLoader helpers (*src/tokenization/*). I then implemented the TF-IDF + Logistic Regression

baseline (*src/training/baseline.py*) and the bidirectional LSTM (Hochreiter & Schmidhuber, 1997) initialised with GloVe 100-dimensional embeddings (*src/models/bilstm.py*). The BiLSTM training loop (*src/training/bilstm.py*) uses Adam with gradient clipping, F1-based checkpointing, and Apple MPS device support. I also hardened the final modular architecture: app services in *src/app/service.py*, inference split across *src/inference/*, and evaluation split across *src/evaluation/*. I produced the presentation outline, 10 acceptance test cases with real model outputs, the submission checklist, and this document.

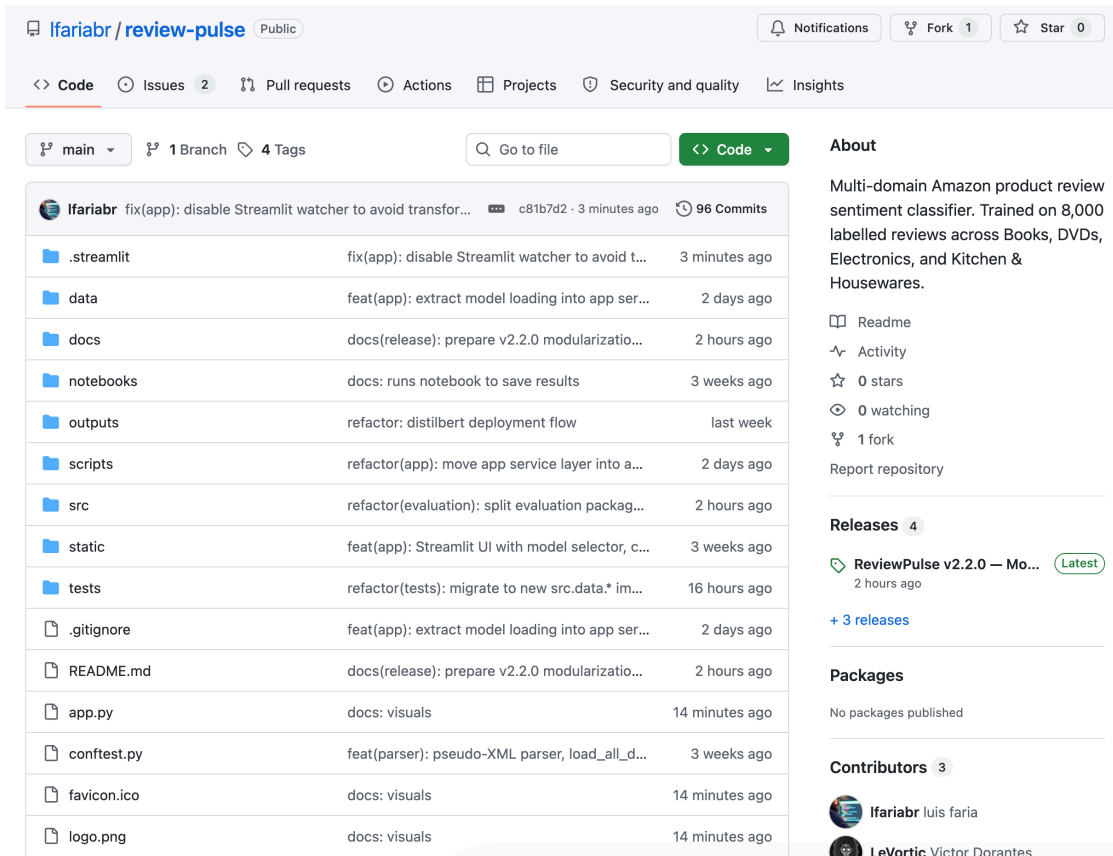
An important ethical consideration is that the Blitzer et al. (2007) dataset uses filename-derived labels rather than human raters. We audited for possible rating/text conflicts and ambiguous boundary cases and found zero ambiguous rows, but the risk remains relevant in broader sentiment classification settings. BiLSTM and DistilBERT confidence values are uncalibrated (high confidence does not imply high accuracy) and both models generalise poorly to out-of-distribution text such as logistics reviews (Bender et al., 2021). Any production deployment requires human oversight and periodic label audits.

I estimate my contribution at 50% as the primary technical implementer and final refactor lead. Victor contributed 25% covering DistilBERT/model experimentation, integration support, dataset analysis, error analysis, and ethics. Samiran contributed 25% through supported problem framing, live demonstration support, and future improvement planning, including review-classification challenges such as negation, sarcasm, review length, and domain differences.

4. Appendices

4.1 Appendix A – Github Repository

<https://github.com/lfariabr/review-pulse>



lfariabr / review-pulse Public

Notifications Fork 1 Star 0

<> Code Issues 2 Pull requests Actions Projects Security and quality Insights

main 1 Branch 4 Tags Go to file Code

File/Folder	Commit Message	Time Ago
.streamlit	fix(app): disable Streamlit watcher to avoid transfor...	3 minutes ago
data	feat(app): extract model loading into app ser...	2 days ago
docs	docs(release): prepare v2.2.0 modularizatio...	2 hours ago
notebooks	docs: runs notebook to save results	3 weeks ago
outputs	refactor: distilbert deployment flow	last week
scripts	refactor(app): move app service layer into a...	2 days ago
src	refactor(evaluation): split evaluation packag...	2 hours ago
static	feat(app): Streamlit UI with model selector, c...	3 weeks ago
tests	refactor(tests): migrate to new src.data.* im...	16 hours ago
.gitignore	feat(app): extract model loading into app ser...	2 days ago
README.md	docs(release): prepare v2.2.0 modularizatio...	2 hours ago
app.py	docs: visuals	14 minutes ago
confest.py	feat(parser): pseudo-XML parser, load_all_d...	3 weeks ago
favicon.ico	docs: visuals	14 minutes ago
logo.png	docs: visuals	14 minutes ago

About
Multi-domain Amazon product review sentiment classifier. Trained on 8,000 labelled reviews across Books, DVDs, Electronics, and Kitchen & Housewares.
Readme Activity 0 stars 0 watching 1 fork Report repository

Releases 4
ReviewPulse v2.2.0 — Mo... Latest 2 hours ago
+ 3 releases

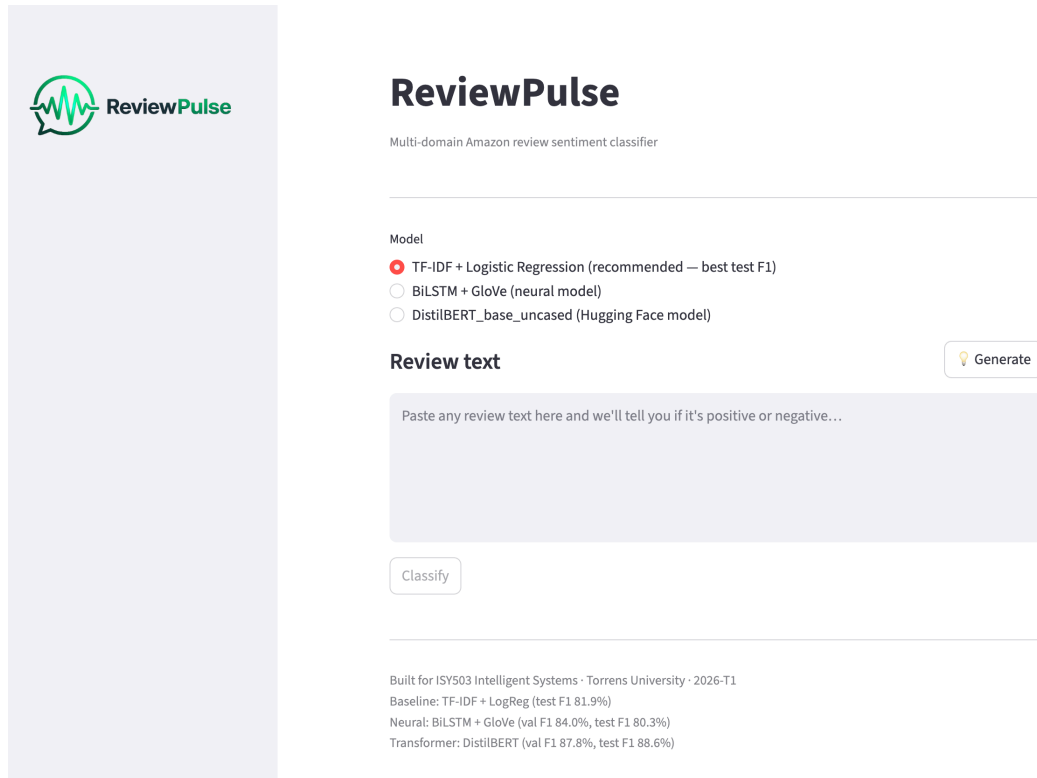
Packages
No packages published

Contributors 3
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Fig A1. Github repository main page.

4.2 Appendix B – Live Application

<https://review-pulse.streamlit.app/>



ReviewPulse
Multi-domain Amazon review sentiment classifier

Model

- ☒ TF-IDF + Logistic Regression (recommended — best test F1)
- ☐ BiLSTM + GloVe (neural model)
- ☐ DistilBERT_base_uncased (Hugging Face model)

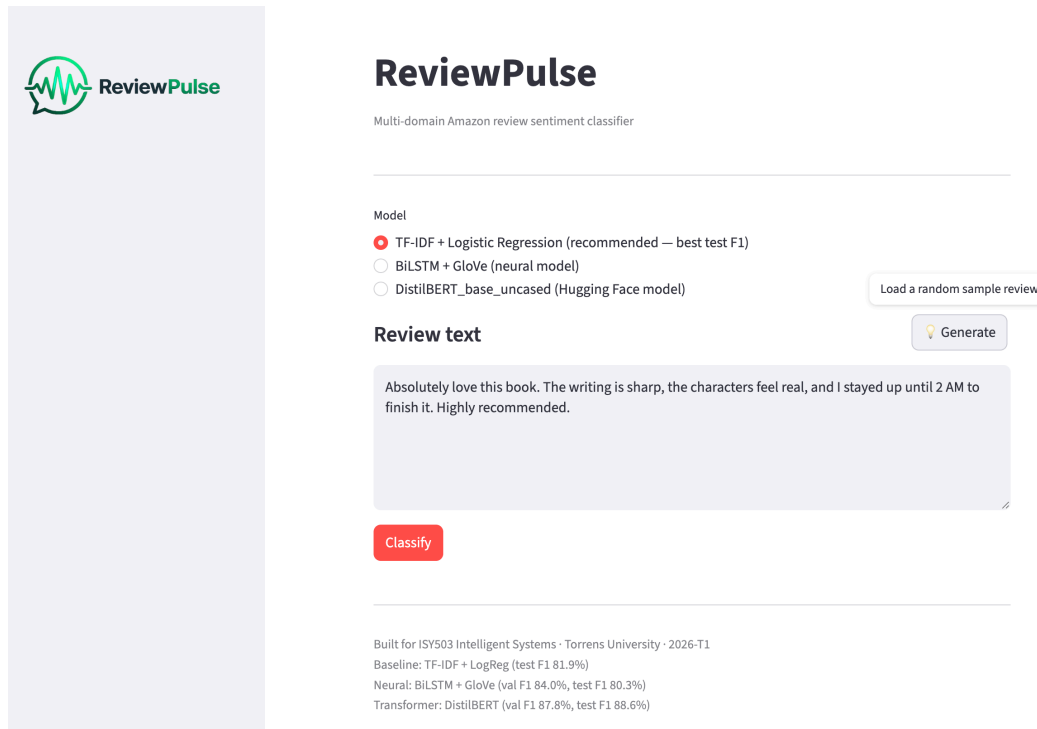
Review text Generate

Paste any review text here and we'll tell you if it's positive or negative...

Classify

Built for ISY503 Intelligent Systems · Torrens University · 2026-T1
 Baseline: TF-IDF + LogReg (test F1 81.9%)
 Neural: BiLSTM + GloVe (val F1 84.0%, test F1 80.3%)
 Transformer: DistilBERT (val F1 87.8%, test F1 88.6%)

Figure A2.1. ReviewPulse app homepage with model selector.



ReviewPulse
Multi-domain Amazon review sentiment classifier

Model

- ☒ TF-IDF + Logistic Regression (recommended — best test F1)
- ☐ BiLSTM + GloVe (neural model)
- ☐ DistilBERT_base_uncased (Hugging Face model)

Load a random sample review

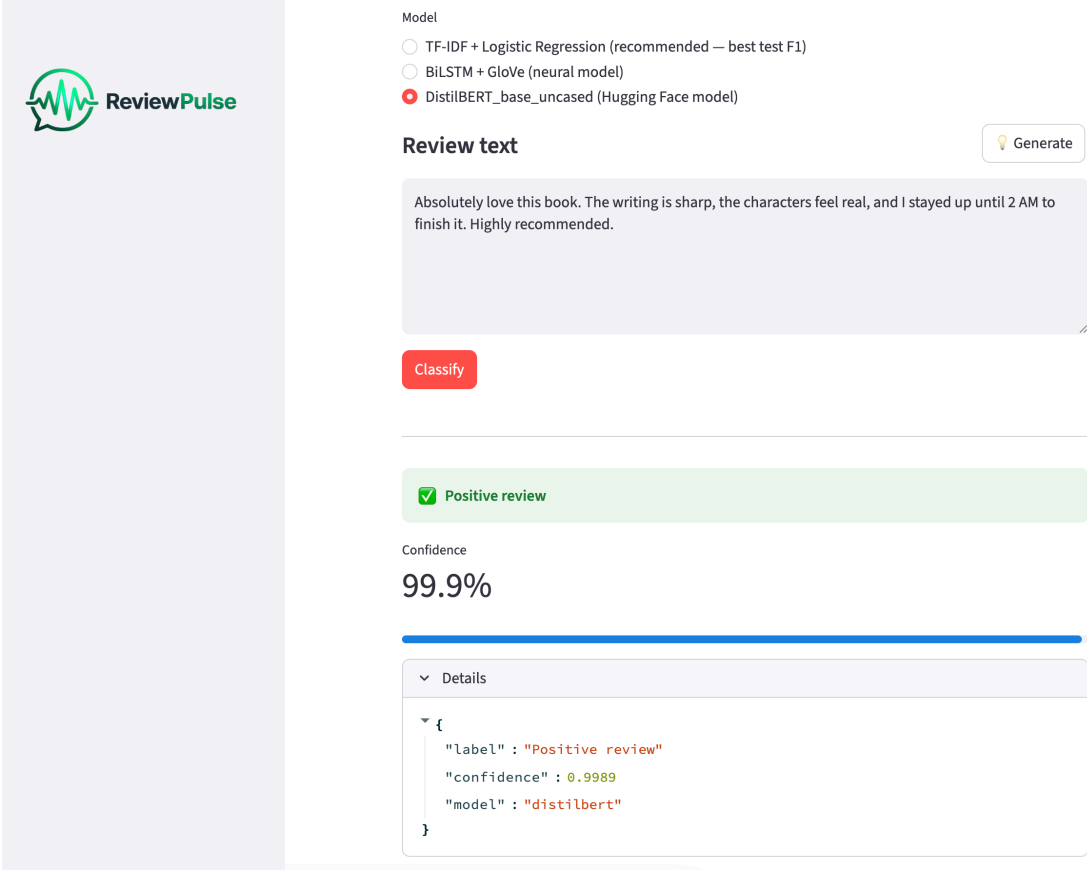
Review text Generate

Absolutely love this book. The writing is sharp, the characters feel real, and I stayed up until 2 AM to finish it. Highly recommended.

Classify

Built for ISY503 Intelligent Systems · Torrens University · 2026-T1
 Baseline: TF-IDF + LogReg (test F1 81.9%)
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Figure A2.2. ReviewPulse app with random sample review generated and ready to classify.



The screenshot shows the ReviewPulse app interface. On the left is a vertical sidebar with the ReviewPulse logo. The main area has a 'Model' section with three radio buttons: 'TF-IDF + Logistic Regression (recommended — best test F1)', 'BiLSTM + GloVe (neural model)', and 'DistilBERT_base_uncased (Hugging Face model)' which is selected. Below this is a 'Review text' input field containing the text: 'Absolutely love this book. The writing is sharp, the characters feel real, and I stayed up until 2 AM to finish it. Highly recommended.' A 'Generate' button is to the right. A red 'Classify' button is below the input. The result is a green bar with a checkmark and the text 'Positive review'. Below this, the 'Confidence' is shown as '99.9%' with a blue progress bar. A 'Details' section is expanded, showing a JSON object:

```
{
  "label": "Positive review",
  "confidence": 0.9989,
  "model": "distilbert"
}
```

Figure A2.3. ReviewPulse app with input review text classified.

3.3 Appendix C – Sample Test Cases Output

Ten acceptance test cases run against the baseline and BiLSTM checkpoints (baseline checkpoint *outputs/baseline.joblib*, BiLSTM checkpoint *outputs/bilstm.pt* epoch 9).

DistilBERT is integrated into the app as the third selectable model using *outputs/distilbert.pt*.

Case	Input (excerpt)	Baseline	BiLSTM
Clear positive	This blender is absolutely incredible...	Positive 73.8%	Positive 97.9%
Clear negative	Broke after two days...	Negative 95.9%	Negative 99.6%
Negation trap	This is not bad at all...	Negative 66.9%	Negative 74.6%
Sarcasm	Oh great, stopped working after a week...	Negative 52.5%	Negative 64.6%
Domain-shifted (books)	One of the best thrillers I have read...	Positive 69.4%	Positive 86.2%

Table A3. Sample Test Cases Output.

Full results: <https://github.com/lfariabr/review-pulse/blob/main/docs/assessment-files/demo-test-cases.md>.

3.4 Appendix D – Future Work

Priority	Extension	Rationale
Shipped	DistilBERT	Victor implemented Hugging Face <i>distilbert-base-uncased</i> ; test F1 88.6%, live in app
High	RoBERTa	Larger pretrained transformer; expect to push F1 beyond 90%
High	Confidence calibration (Platt scaling)	BiLSTM logits are uncalibrated; 98% confidence $\neq$ 98% accuracy
Medium	Additional training domains	Current model is trained on 4 product categories only; broader domains would validate generalisation
Medium	LIME / attention visualisation	Makes predictions auditable - necessary for any production or high-stakes deployment
Low	FastAPI backend	Decouple inference from Streamlit for easier integration with external systems

Table A4. Future Work possibilities and opportunities.

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (GPT-5.4)
- Anthropic Claude Opus 4.6

Both tools were used to assist with understanding ML concepts, structuring the technical manual, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"I built a BiLSTM sentiment classifier with pack_padded_sequence to stop the LSTM processing padding tokens. My training achieves val F1=84.0% at epoch 9, but test F1 drops to 80.3% while the TF-IDF baseline reaches 81.9% on the same held-out set. Can you help me understand why the neural model didn't generalise better despite the stronger validation score?"*
2. *"I need to compare two approaches for sentiment analysis – TF-IDF + Logistic Regression vs BiLSTM with GloVe – in an academic report where the simpler model wins on the held-out test set. How do I frame this honestly without undermining the neural architecture requirement of the assessment rubric?"*
3. *"Format this as APA 7th: Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, 2021, article titled On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?, conference ACM FAccT 2021, pages 610 to 623."*

I confirm that the use of these tools has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

5. References

- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM FAccT Conference* (pp. 610–623). <https://doi.org/10.1145/3442188.3445922>
- Blitzer, J., Dredze, M., & Pereira, F. (2007). Biographies, Bollywood, Boom-boxes and Blenders: Domain adaptation for sentiment classification. In *Proceedings of the 45th Annual Meeting of the ACL* (pp. 440–447). ACL. <https://aclanthology.org/P07-1056/>
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9 (8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>